Nut intake and adiposity: meta-analysis of clinical trials.

BACKGROUND: Epidemiologic studies have shown an inverse association between the frequency of nut consumption and body mass index (BMI) and risk of obesity. However, clinical trials that evaluated nut consumption on adiposity have been scarce and inconclusive. OBJECTIVE: We performed a systematic review and meta-analysis of published, randomized nut-feeding trials to estimate the effect of nut consumption on adiposity measures. DESIGN: MEDLINE and the Cochrane Central Register of Controlled Trials databases were searched for relevant clinical trials of nut intake that provided outcomes of body weight, BMI (in kg/m(2)), or waist-circumference measures and were published before December 2012. There were no language restrictions. Two investigators independently selected and reviewed eligible studies. The weighted mean difference (WMD) between nut or control diets was estimated by using a random-effects meta-analysis with 95% CIs. RESULTS: Thirty-three clinical trials met our inclusion criteria. Pooled results indicated a nonsignificant effect on body weight (WMD: -0.47 kg; 95% CI: -1.17, 0.22 kg; I(2) = 7%), BMI (WMD: -0.40 kg/m(2); 95% CI: -0.97, 0.17 kg/m(2); I(2) = 49%), or waist circumference (WMD: -1.25 cm; 95% CI: -2.82, 0.31 cm; I(2) = 28%) of diets including nuts compared with control diets. These findings were remarkably robust in the sensitivity analysis. No publication bias was shown. CONCLUSION: Compared with control diets, diets enriched with nuts did not increase body weight, body mass index, or waist circumference in controlled clinical trials.